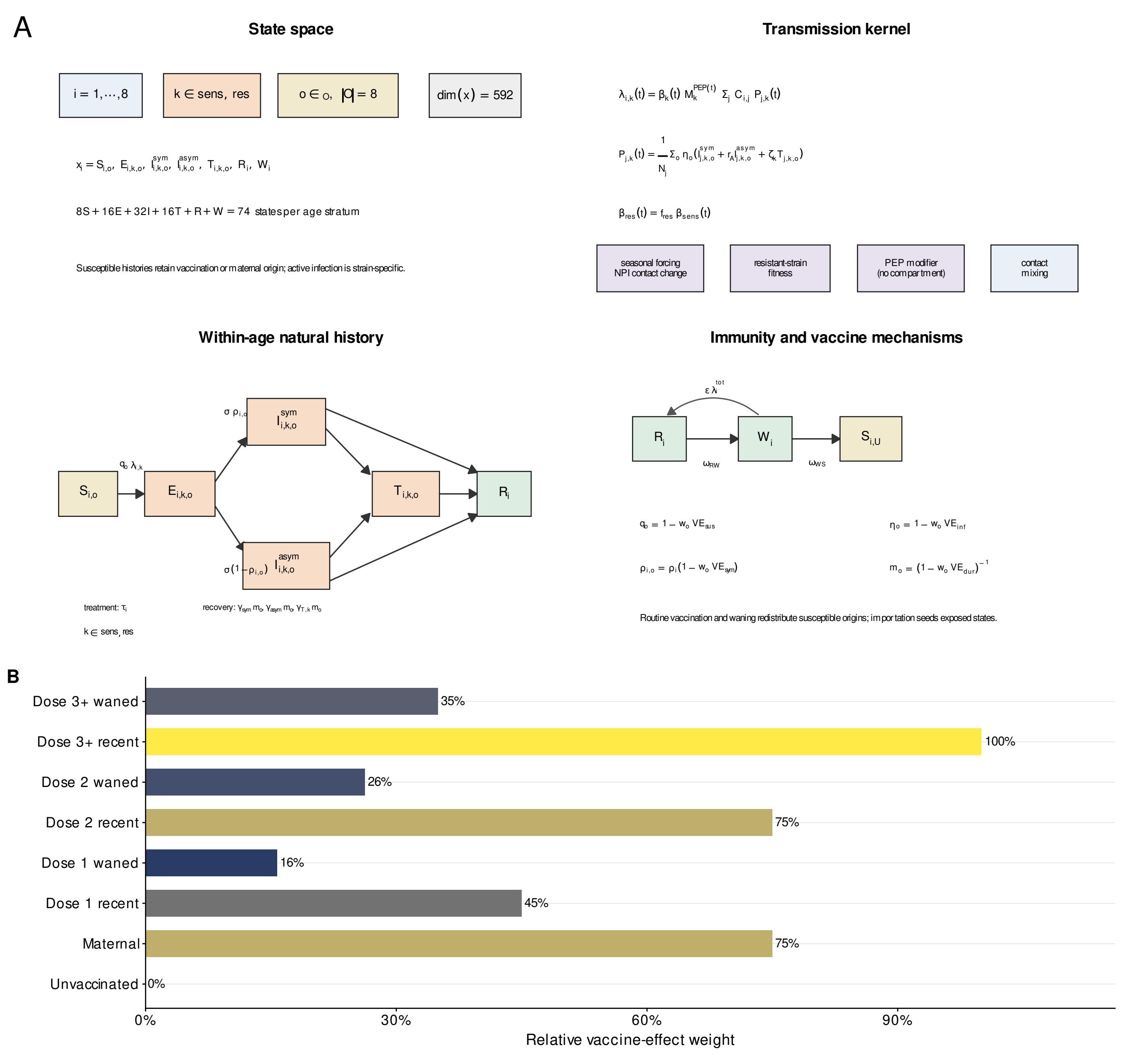
A
State space
i = 1,…,8
k ∈ sens, res
o ∈ O, |O| = 8
dim(x) = 592
x_i = S_{i,o}, E_{i,k,o}, I^{sym}_{i,k,o}, I^{asym}_{i,k,o}, T_{i,k,o}, R_i, W_i
8S + 16E + 32I + 16T + R + W = 74 states per age stratum
Susceptible histories retain vaccination or maternal origin; active infection is strain-specific.
Transmission kernel
λ_{i,k}(t) = β_k(t) M_k^{PEP(t)} Σ_j C_{i,j} P_{j,k}(t)
P_{j,k}(t) = 1/N_j Σ_o η_o (I^{sym}_{j,k,o} + r_A I^{asym}_{j,k,o} + ζ_k T_{j,k,o})
β_res(t) = f_res β_sens(t)
seasonal forcing
NPI contact change
resistant-strain fitness
PEP modifier (no compartment)
contact mixing
Within-age natural history
S_{i,o}
q_o λ_{i,k}
E_{i,k,o}
σ ρ_{i,o}
I^{sym}_{i,k,o}
σ(1 − ρ_{i,o})
I^{asym}_{i,k,o}
T_{i,k,o}
R_i
treatment: τ_i
recovery: γ_sym m_o, γ_asym m_o, γ_{T,k} m_o
k ∈ sens, res
Immunity and vaccine mechanisms
ε λ^{tot}_i
R_i
W_i
S_{i,U}
ω_RW
ω_WS
q_o = 1 − w_o VE_sus
η_o = 1 − w_o VE_inf
ρ_{i,o} = ρ_i (1 − w_o VE_sym)
m_o = (1 − w_o VE_dur)^{−1}
Routine vaccination and waning redistribute susceptible origins; importation seeds exposed states.
B
Dose 3+ waned
35%
Dose 3+ recent
100%
Dose 2 waned
26%
Dose 2 recent
75%
Dose 1 waned
16%
Dose 1 recent
45%
Maternal
75%
Unvaccinated
0%
0%
30%
60%
90%
Relative vaccine-effect weight